

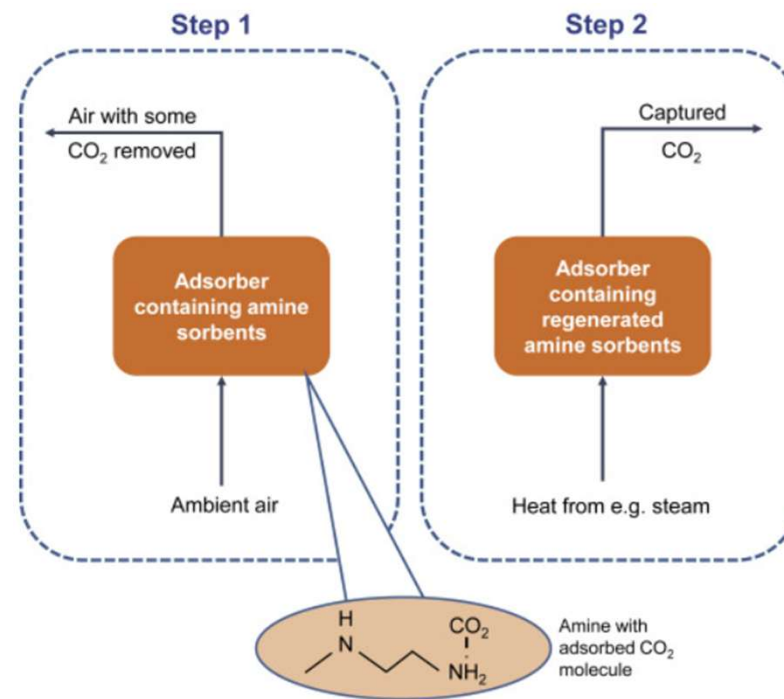
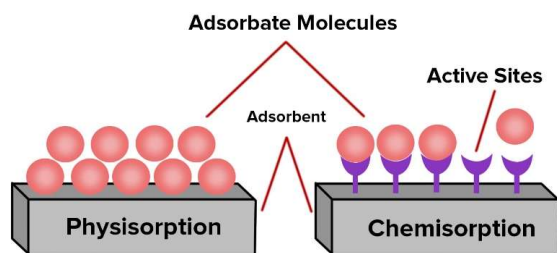
DAC using solid sorbents

Solid-based DAC:

Porous materials like **metal-organic frameworks**, zeolites, porous silica, etc.

Solid sorbents can remove CO_2 from gas mixtures via weak molecular forces called **physisorption** or strong covalent bonding called **chemisorption**.

Solid-supported amines have received significant attention for DAC due to their ability in CO_2 capturing and low-temperature regeneration conditions (**~ 50 to 120°C**)

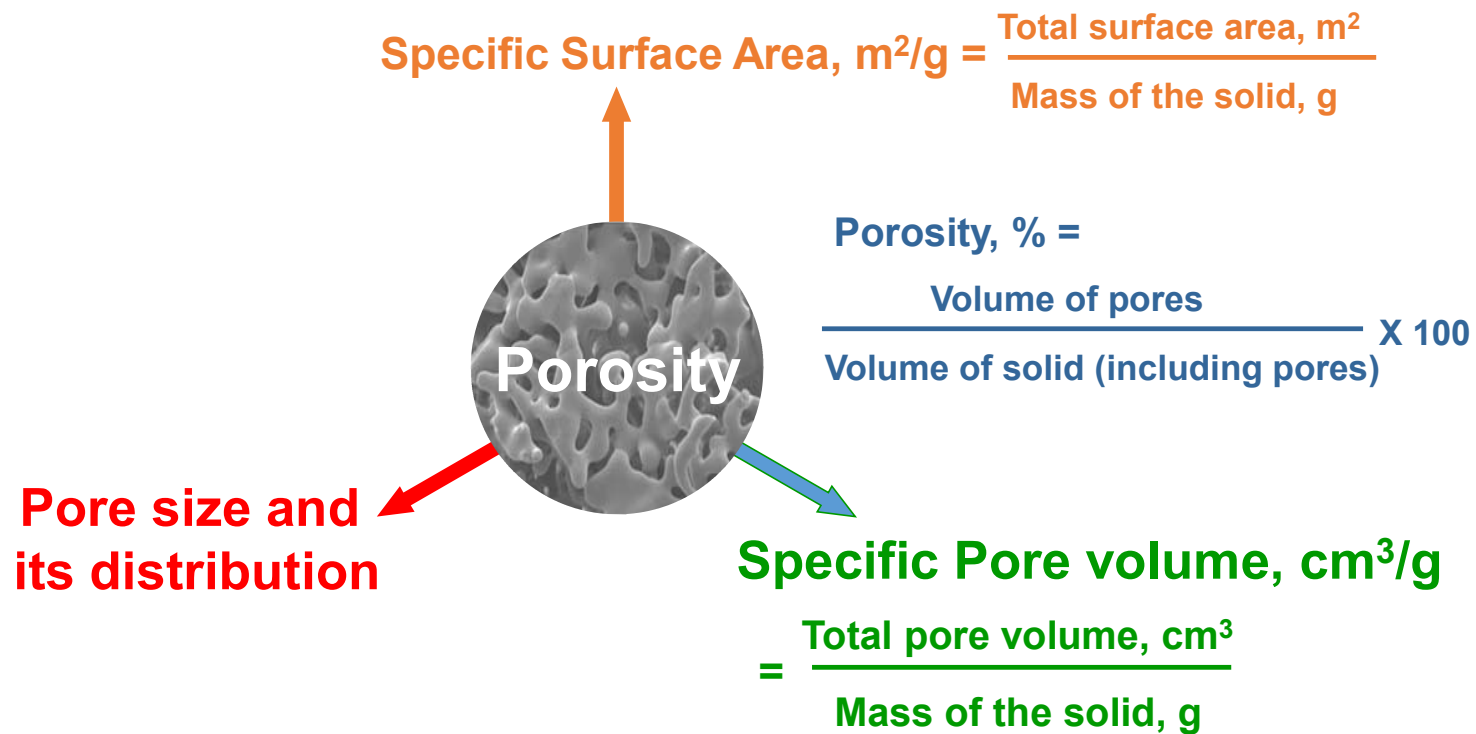


Amine-functionalized solid sorbents

Measure of Porosity

There are three parameters used as a measure of porosity;

- specific surface area,
- specific pore volume or porosity
- pore size and its distribution.



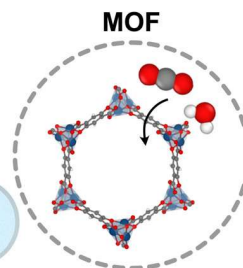
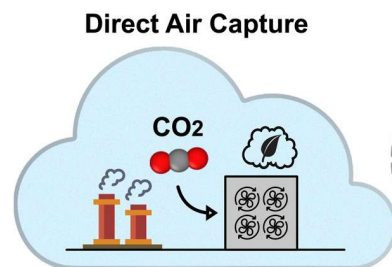
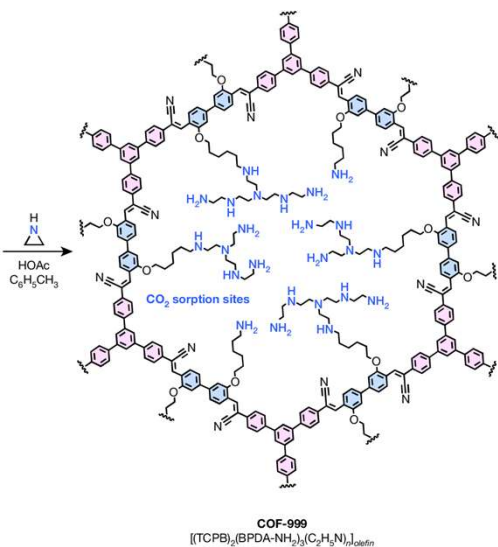
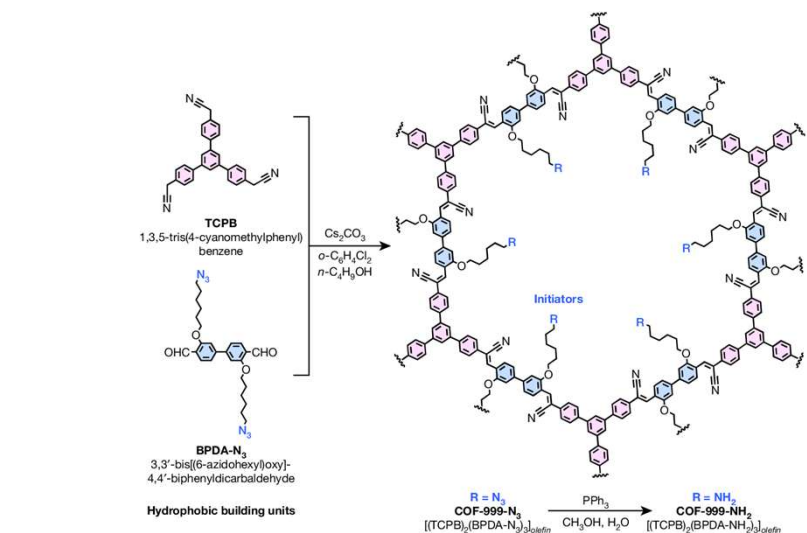
Size of Pores (IUPAC Standard)



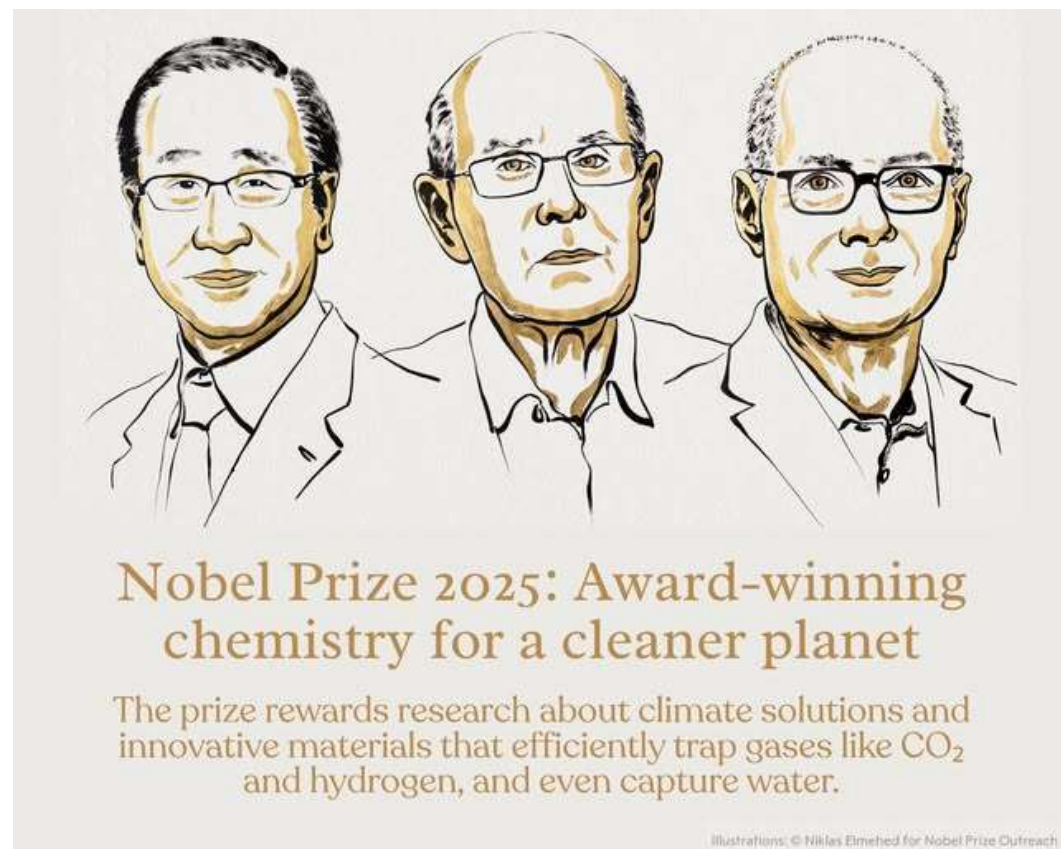
Porous material are classified according to the size of pores: material with pores less than 2 nm are called micropores, materials with pores between 2 and 50 nm are called mesopores, and material with pores greater than 50 nm are macropores

Sing, K. S. W. et al. Reporting Physisorption Data for Gas/Solid Systems. Pure & Appl. Chem. 57, 603-619 (1985).

PHYSISORPTION	CHEMISORPTION
WEAK, LONG-RANGE BONDING van der Waals interactions	STRONG, SHORT-RANGE BONDING Chemical bonding involved.
NOT SURFACE SPECIFIC Physisorption takes place between all molecules on any surface providing the temperature is low enough.	SURFACE SPECIFIC E.g. Chemisorption of hydrogen takes place on transition metals but not on gold or mercury.
$\Delta H_{\text{ads}} = 5 \dots\dots 50 \text{ kJ mol}^{-1}$	$\Delta H_{\text{ads}} = 50 \dots\dots 500 \text{ kJ mol}^{-1}$
Non activated with equilibrium achieved relatively quickly. Increasing temperature always reduces surface coverage.	Can be activated, in which case equilibrium can be slow and increasing temperature can favour adsorption.
No surface reactions.	Surface reactions may take place:- Dissociation, reconstruction, catalysis.

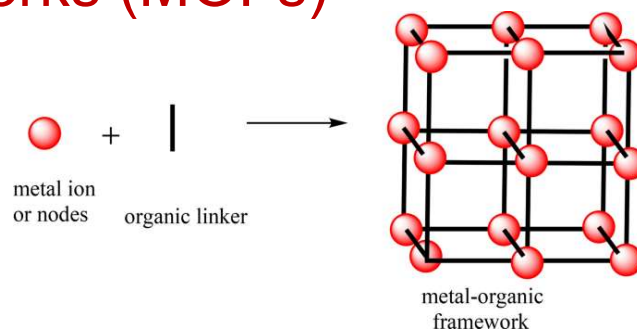


2025 Nobel Prize in Chemistry went to Susumu Kitagawa, Richard Robson and Omar Yaghi for developing metal-organic frameworks (MOFs)



Metal-Organic Frameworks (MOFs)

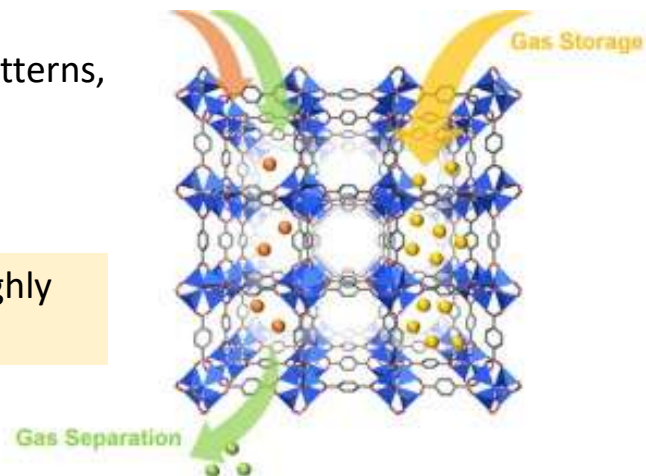
MOFs are porous materials with up to 94% empty space in their framework.



Metal-Organic Frameworks (MOFs) are advanced porous materials with ultra-high surface areas and tunable pore sizes that make them highly effective for gas storage and separation.

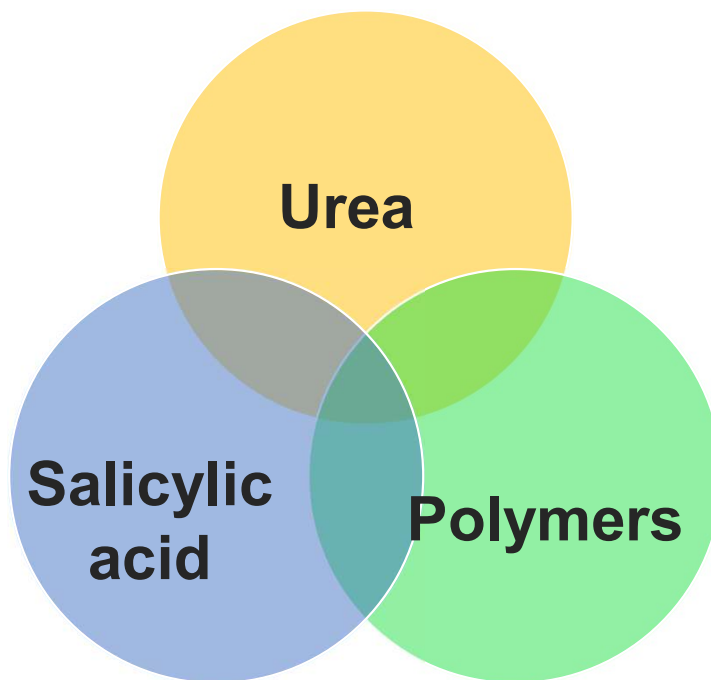
Building blocks self-assemble into repeating, grid-like (reticular) patterns, creating structures with tiny, interconnected cavities or pores.

The size and chemical nature of the pores can be designed to be highly specific, allowing MOFs to selectively adsorb certain molecules.



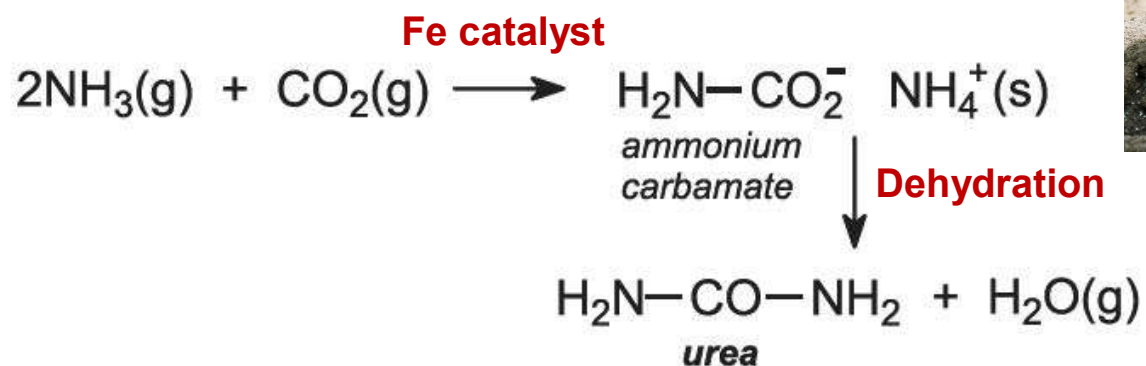
CO₂ use for chemicals synthesis

Annually, about 130 Mt of CO₂ are used to manufacture **urea, salicylic acid, and polycarbonates**, among which the urea process consumes most of the CO₂ industrially.



Urea synthesis using CO₂

- Urea is synthesized from CO₂ and NH₃.
- The reaction generally operates at 150-250 °C and at a pressure of 2 MPa.
- It is mainly used in the production of fertilizers.

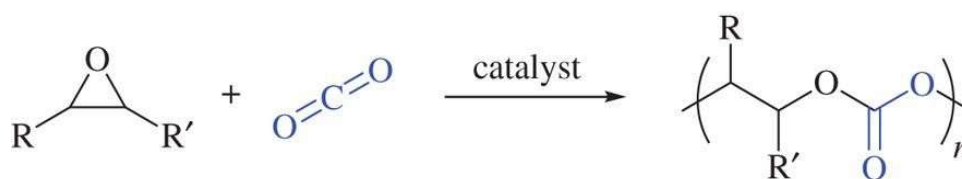


Polycarbonates from CO₂

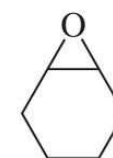


Polycarbonates synthesis using CO₂

- Polycarbonates are synthesized via ring-opening copolymerization of epoxides with carbon dioxide.
- ZnEt₂ is a catalyst.
- Examples: plastic lenses in eyewear, automotive components, protective gear, greenhouses, digital Disks (CDs, DVDs, and Blu-ray).



common epoxides



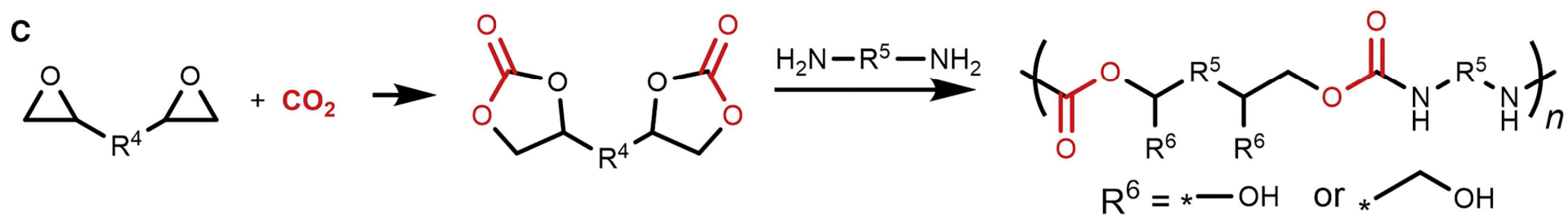
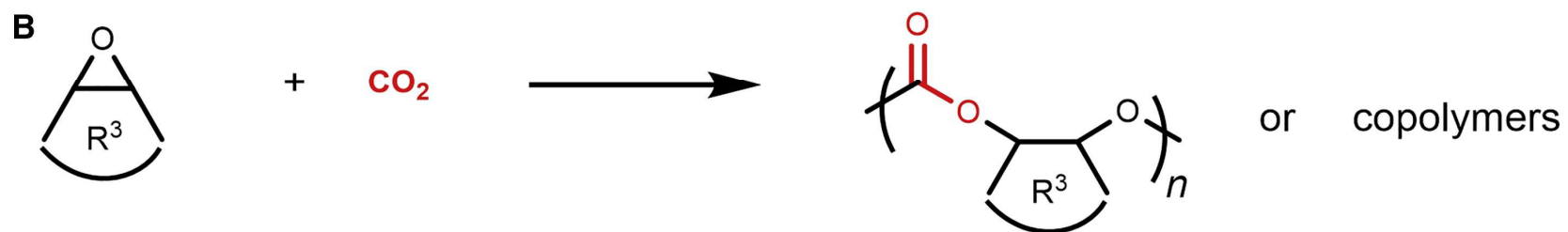
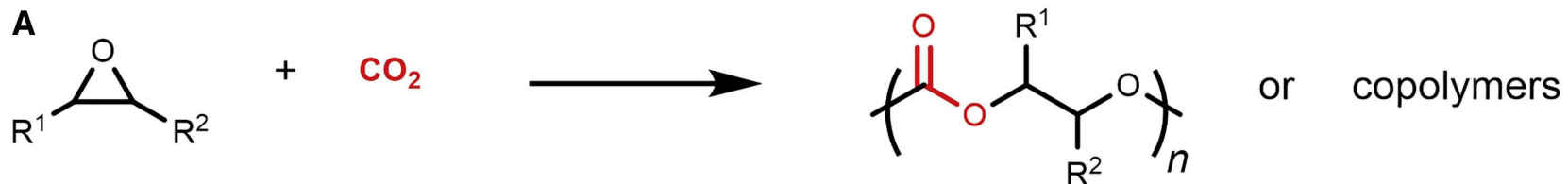
CHO

Cyclohexene
epoxide



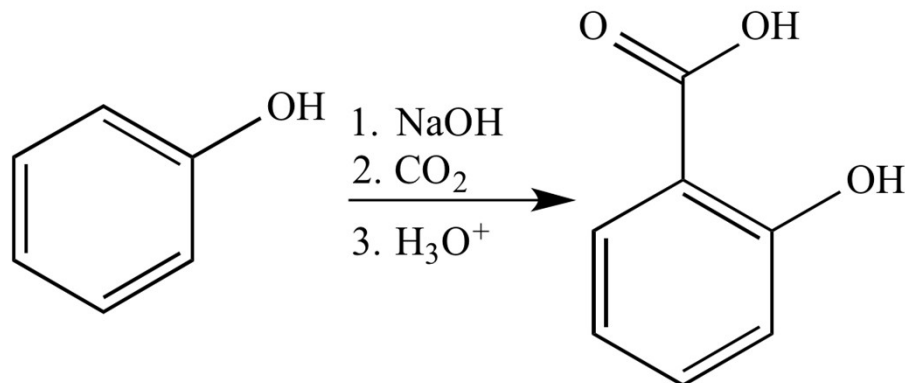
PO

Propylene
epoxide



Salicylic acid synthesis using CO₂

- Salicylic acid is used **to treat skin disorders**.
- It is produced commercially **via the Kolbe-Schmitt process**.
- Phenol and sodium hydroxide are reacted to form sodium phenoxide. Then phenoxide reacts with CO₂ to form sodium salicylate. The salicylate is acidified to give salicylic acid.



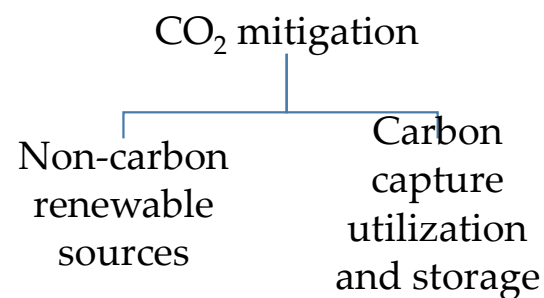
Phenol

Salicylic acid

Global carbon budget



- Temperature rise by year 2100 will be > 4 °C
- Goal is to mitigate below 1.5 – 2 °C
- Point Sources 5 – 20%
- Direct Air capture – 420 ppm (0.042 %)



Importance of Clean CO₂ capture units

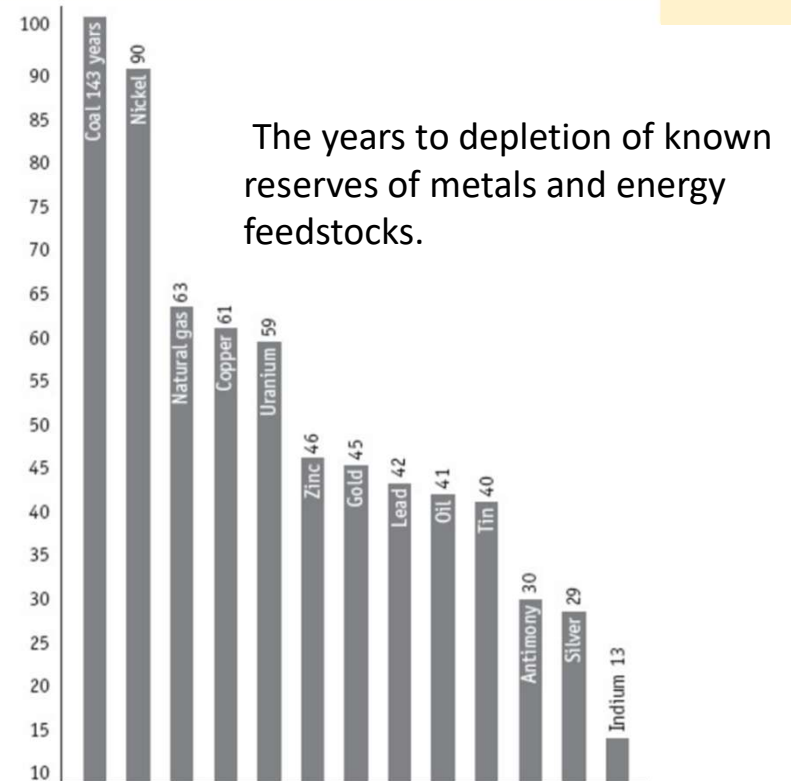
- Globally 100 plants are either in pilot or working stage for CO₂ capture
- To meet the Paris agreement, focus should be needed on lean carbon sources with < 8 % of CO₂ sources
- Currently 18 plants operating worldwide for lean carbon capture in small scale
- DAC is the only negative emission technology
- Clean CO₂ capture is currently receiving increased levels of attention and monetary support from both public and private sector organizations

Ritchie, H et.al., *Our World Data* 2020

Global Energy Review: CO₂ Emissions in 2021

Sustainable Consumption and Production

Today it has become clear that our world urgently needs to adopt more sustainable consumption and production (SCP) patterns to reduce the use of natural resources and the release of carbon dioxide (CO₂) while increasing the recycling rate for manufactured goods. Since the Industrial Revolution, the world has experienced economic growth unprecedented in human history

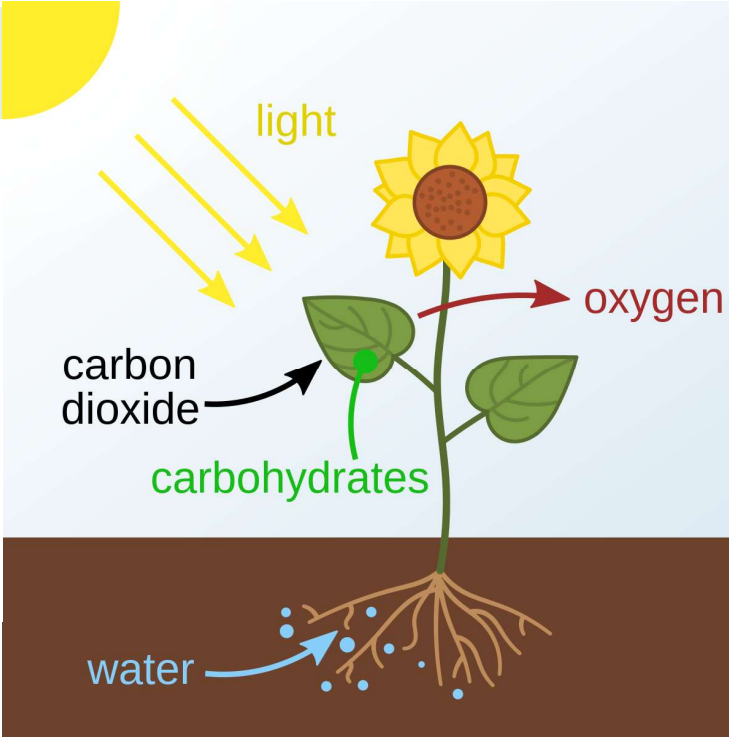
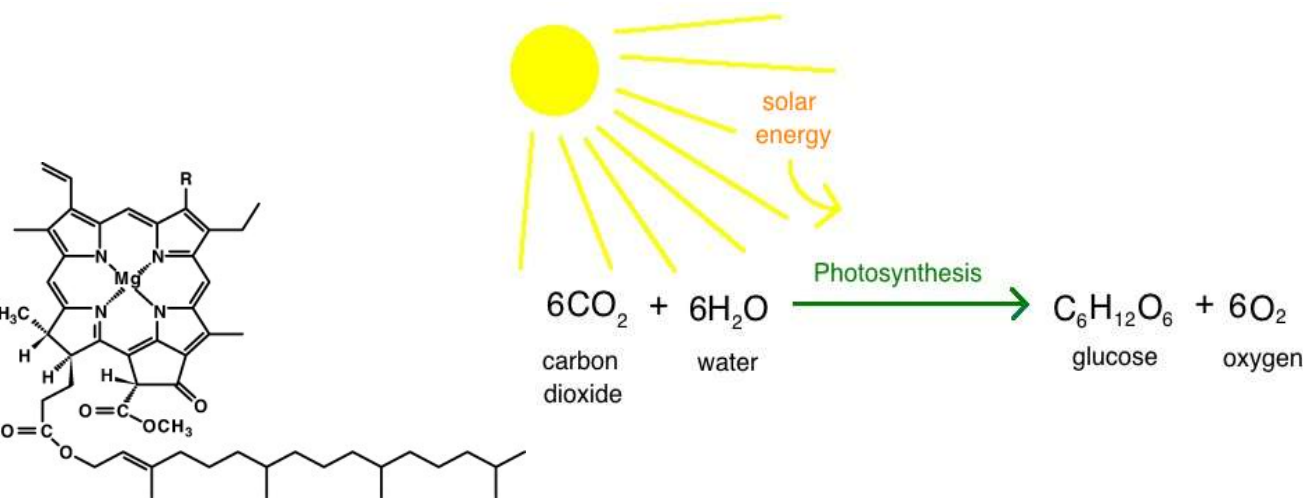


What is Environment?

Many environmental problems and processes are chemical, and to understand them, basic chemistry involved is necessary

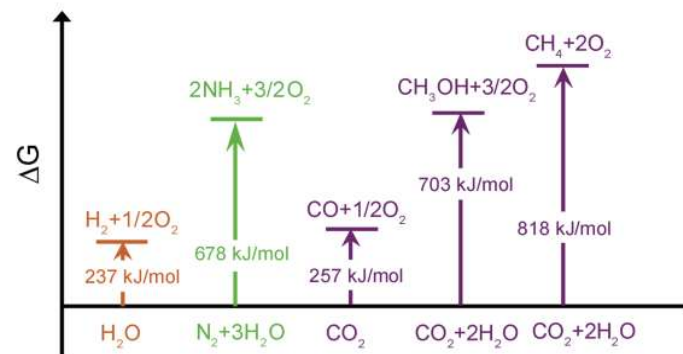
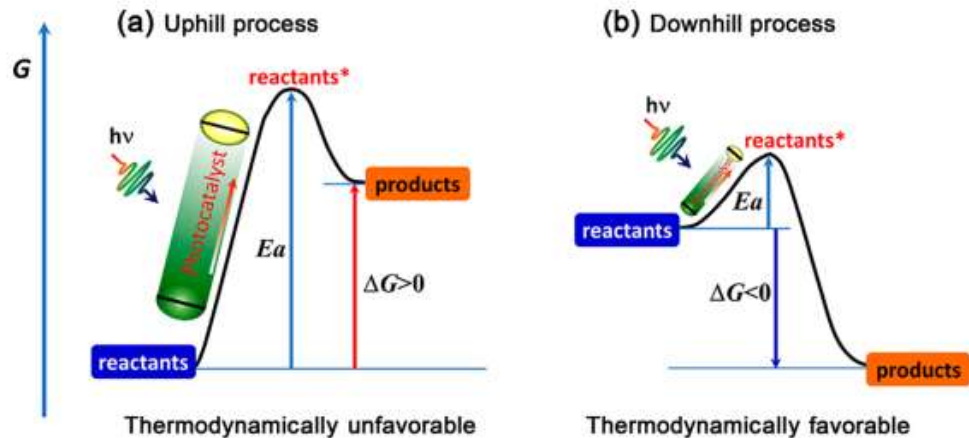
Environment : Atmosphere-water-earth surroundings that make life possible

Biological process of photosynthesis created the oxygen atmosphere in which we live in



To Drive uphill reactions

Broad definition by IUPAC on photocatalyst as “catalyst able to produce, upon absorption of light, chemical transformations of the reaction partners...”

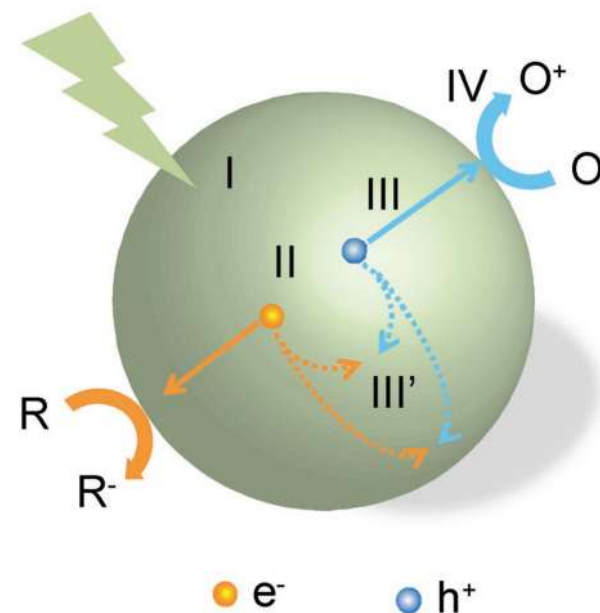


Energy diagrams of representative thermodynamically uphill reactions.

How does a photocatalyst work?

- (I) light absorption to generate electron-hole pairs;
- (II) separation of excited charges;
- (III) transfer of electrons and holes to the surface of photocatalysts;
- and
- (IV) utilization of charges on the surface for redox reactions

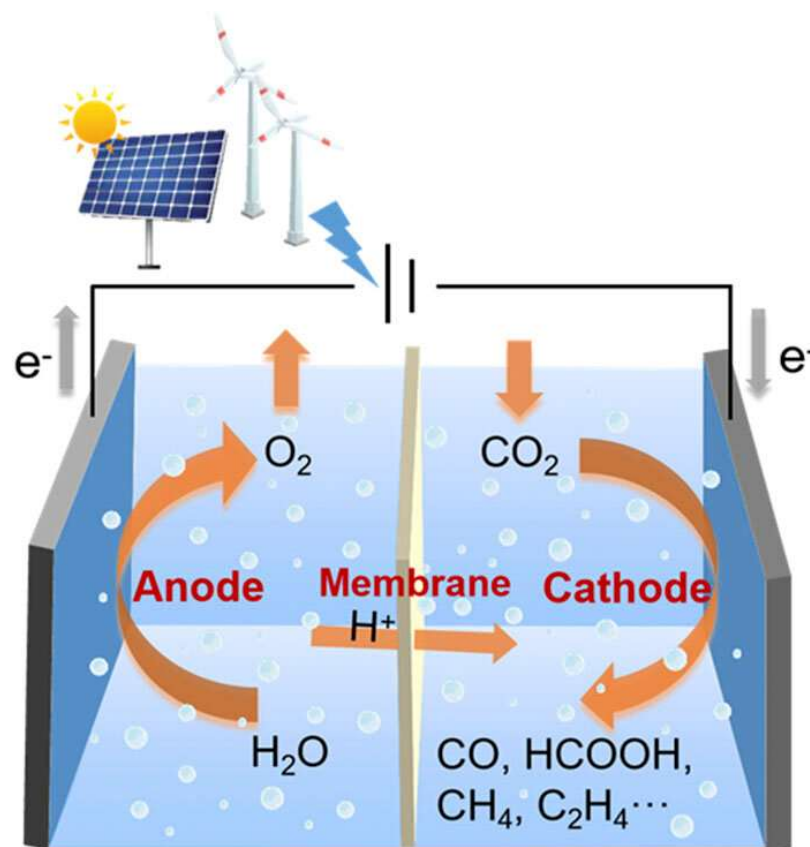
The recombination dissipates the harvested energy in the form of heat (nonradiative recombination) or light emission (radiative recombination).



The long-lived photogenerated charges on the surface have the potential to promote different redox reactions

Electrochemical reduction of CO₂

Develop efficient electrocatalysts capable of reducing CO₂ to valuable hydrocarbon products at a low overpotential with high selectivity and stability.



POPULATION AND THE SUSTAINABLE DEVELOPMENT GOALS: **The UN 17 Sustainable Goals**

Aim: shared blueprint for peace and prosperity for people and the planet, now and into the future

Global partnership among developing and developed countries

“Development that meets the needs of the present Without compromising the ability of the future generations to meet their own needs”

poverty, inequality, climate change, environmental degradation, peace and justice



Source Reduction

- By reducing or eliminating the generation of harmful and persistent wastes, waste minimization supports efforts to promote a more sustainable society.
- Waste minimization involves redesigning products and processes and/or changing societal patterns of consumption and production

How to do it?

- The best way to deal with trash is to not have any!
- Reducing the amount of trash you have to throw out actually prevents waste from piling up in the first place.
- To reduce your waste, avoid unnecessary packaging and items designed to be used only once.
 - Reduce the need for 'single use' plastic bags by bringing your own bags when you shop and use a travel mug when you buy coffee.
 - Choose durable, reusable products to make less trash.

Fiber optic communication



Reuse

- Reusing items can save energy and money and prolong the item's useful life.
- Extend the life of items you buy by reusing them.
 - For example, reuse containers and jars, and donate still usable household goods and clothing to charity.



Minimizing Waste at IIT Hyderabad

IIT Hyderabad is dedicated to minimizing waste, maximizing resource efficiency, and implementing sustainable disposal practices. The Institute's waste management initiatives focus on environmental responsibility and innovative solutions.

1. Wastewater Processing

IITH has established an advanced wastewater treatment plant on campus to recycle and reuse wastewater. The treated water is repurposed for cooling systems and flush water, significantly reducing freshwater consumption and promoting sustainable water management.

2. Solid Waste Recycling

A comprehensive recycling program efficiently manages various waste materials, including packing materials, paper, cardboard, scrap metal, and hazardous e-waste. The program encourages campus residents to segregate waste at the source, ensuring efficient recycling and minimizing landfill waste.

3. Biogas Production

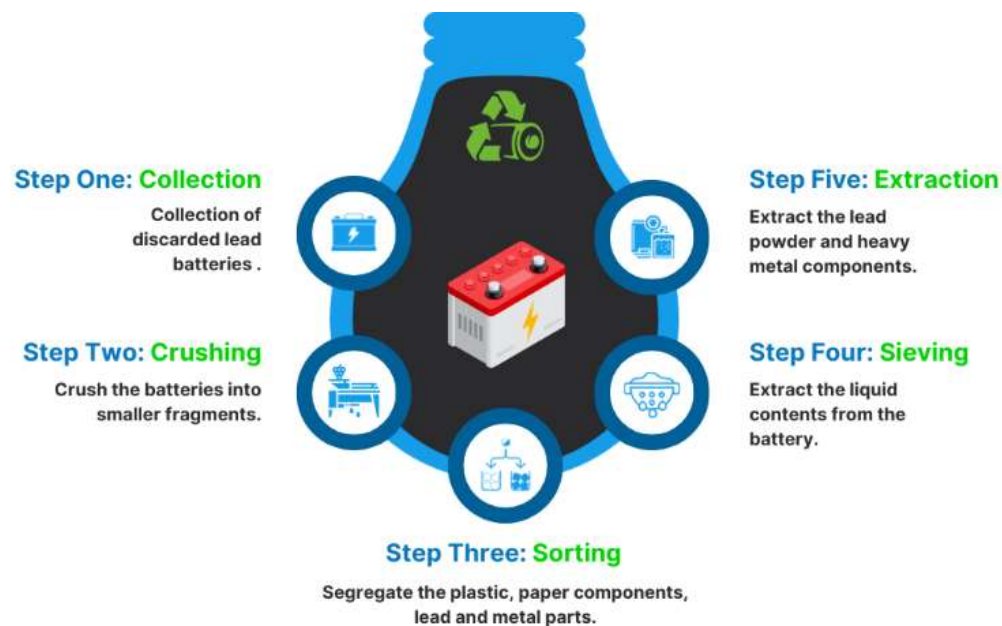
The campus processes wet waste in a biodigester to produce biogas, which serves as cooking fuel for the student dining hall. Food residues and vegetable peelings from dining areas are utilized as feedstock, significantly reducing fossil fuel consumption. The byproduct, biosolid waste, is converted into nutrient-rich compost at the Resource Recovery Park, further supporting campus greenery.

By integrating these waste management strategies, IIT Hyderabad minimizes its environmental impact while serving as a model for sustainable practices, promoting a circular economy, and fostering eco-consciousness within its community.

Recycle

- Every day we use products made from recycled materials.
- Take your glass, cans, newspapers, milk jugs and other acceptable recyclable items to your local transfer station, drop off location or place out for curbside collection so that they can be turned into new products like jackets
- Frisbees, paper products, and soda cans. Recycling saves money, energy, and the environment.

Pb-acid battery Recycling



Heroes of sustainability

- The 2004 Nobel peace prize was awarded to Kenyan environmentalist Wangari Maathai
- This is the greatest recognition given to the cause of environment at international level.
- Maathai Kenya's Deputy Environment Minister and Founder of Kenya based Green Belt Movement.
- This movement planted about 30 million trees across Africa.
- This has helped in slowing desertification, preserving forest for wildlife and food for future generation.
- 5th June is the world environment day.



Wangari Maathai

Indian environmentalists



Rajender Singh

**Water
conservation**

“Waterman of India” Ramon
Magsaysay awardee in 2001

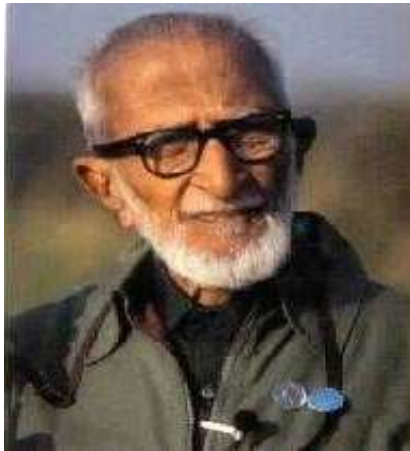


M. C. Metha

**Green
advocate**

Ramon Magsaysay awardee in 1997
Goldman Environmental Prize in 1996

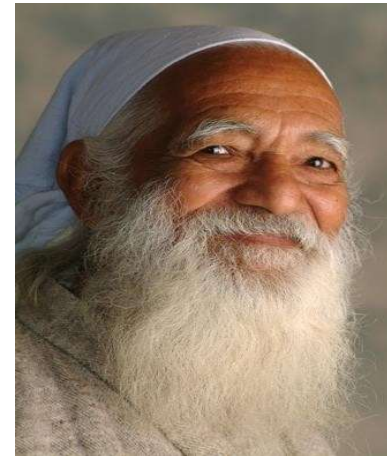
Indian environmentalists



Salim
Ali

Ornithologist

“Birdman of India”
Padma Vibhushan award
in 1976



Sunderlal
Bahuguna

Chipko movement
(forest conservation)

Padma Vibhushan
awardee in 2009

Protect of our Planet Earth

- ✓ Conserve resources
- ✓ Conserve Energy
- ✓ Grow more trees
- ✓ 'RRR' – waste
- ✓ Ban Polyethylene
- ✓ Use Renewables

